SinCos, etc.

$$\cos(x) = \frac{e^{jx} + e^{-jx}}{2}$$
$$\sin(x) = \frac{e^{jx} - e^{-jx}}{2j}$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{N}$$

LTI

Linear: $\alpha_1 x_1(t) + \alpha_2 x_2(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t) \; \forall \; \alpha_1, \alpha_2$
Time invariant: $x(t) \to y(t) \implies x(t - t_0) \to y(t - t_0)$

Convolution

Convolution:

$$y = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

Impulse Response: $h = S[\delta[n]] = S(\delta(t)) = \frac{d}{dt}S(u(t))$

Properties

$x * h = h * x$
$x * (h_1 + h_2) = x * h_1 + x * h_2$
$x * (h_1 * h_2) = (x * h_1) * h_2$

Classifications

Memoryless: $h[k] = 0 = h(t) \; \forall \; k, t \neq 0$
Invertible: $\exists h_I | h * h_I = \delta$
Causal: $h[k] = h(t) = 0 \; \forall \; k, t < 0$
Stable: $\sum_{-\infty}^{\infty} |h[k]| < \infty$,$\int_{-\infty}^{\infty} h(\tau)d\tau < \infty$

Fourier

$$y(t) = e^{j\omega t} * h(t) = e^{j\omega t} \int_{-\infty}^{\infty} h(\tau)e^{-j\omega\tau}d\tau$$
$$= e^{j\omega t}|H(j\omega)|e^{j\underline{/H}(j\omega)}$$

$$a_k = \frac{1}{T}\int_{<T>} x(t)e^{-jk\omega_0 t}dt = \frac{1}{N}\sum_{n=<N>} x[n]e^{-jk\frac{2\pi}{N}n}$$

$$x = \sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=<N>} a_k x_k[n]$$

Real $x(t)$: $a_k = a_{-k}^*$
Even $x(t)$: $a_k = a_{-k}$

Property	**Time Domain**		**F Domain**
Linearity	$Ax_1(t) + Bx_2(t)$	$\overset{FS}{\leftrightarrow}$	$Aa_k + Bb_k$
Even Symmetry	$x(t)$ even	$\overset{FS}{\leftrightarrow}$	a_k even
Odd Symmetry	$x(t)$ odd	$\overset{FS}{\leftrightarrow}$	a_k odd
Time Shifting	$x(t - t_0)$	$\overset{FS}{\leftrightarrow}$	$a_k e^{-jk\omega_0 t_0}$
Frequency Shifting	$x(t)e^{jn\omega_0 t}$	$\overset{FS}{\leftrightarrow}$	a_{k-n}
Time Reversal	$x(-t)$	$\overset{FS}{\leftrightarrow}$	a_{-k}
Conjugation	$x^*(t)$	$\overset{FS}{\leftrightarrow}$	a_{-k}^*
Periodic Convolution	$(x * y)(t)$	$\overset{FS}{\leftrightarrow}$	$a_k b_k$
Multiplication	$x(t)y(t)$	$\overset{FS}{\leftrightarrow}$	$\sum_{n=-\infty}^{\infty} a_n b_{k-n}$
Differentiation	$\frac{d}{dt}x(t)$	$\overset{FS}{\leftrightarrow}$	$jk\omega_0 a_k$
Integration	$\int x(t)dt$	$\overset{FS}{\leftrightarrow}$	$\frac{a_k}{jk\omega_0}$ if $a_0 = 0$)
Parseval's Theorem	$\frac{1}{T}\int_T \|x(t)\|^2 dt$	$\overset{FS}{\leftrightarrow}$	$\sum_{k=-\infty}^{\infty} \|a_k\|^2$

Property	**Time Domain**		**F Domain**
Linearity	$A\,x_1[n] + B\,x_2[n]$	$\overset{DTFS}{\leftrightarrow}$	$A\,a_k + B\,b_k$
Even Symmetry	$x[n]$ even	$\overset{DTFS}{\leftrightarrow}$	a_k even
Odd Symmetry	$x[n]$ odd	$\overset{DTFS}{\leftrightarrow}$	a_k odd
Time Shifting	$x[n - n_0]$	$\overset{DTFS}{\leftrightarrow}$	$a_k\, e^{-j\frac{2\pi}{N}kn_0}$
Frequency Shifting	$x[n]\, e^{j\frac{2\pi}{N}n_0 n}$	$\overset{DTFS}{\leftrightarrow}$	$a_{(k-n_0) \bmod N}$
Time Reversal	$x[-n]$	$\overset{DTFS}{\leftrightarrow}$	a_{-k} (index $\%N$
Conjugation	$x^*[n]$	$\overset{DTFS}{\leftrightarrow}$	a_{-k}^*
Multiplication	$x[n]\, y[n]$	$\overset{DTFS}{\leftrightarrow}$	$\frac{1}{N}\sum_{m=0}^{N-1} a_m\, b_{(k}$
Difference	$x[n] - x[n-1]$	$\overset{DTFS}{\leftrightarrow}$	$a_k\left(1 - e^{-j\frac{2\pi}{N}k}\right)$
Parseval's Theorem	$\frac{1}{N}\sum_{n=0}^{N-1} \|x[n]\|^2$	$\overset{DTFS}{\leftrightarrow}$	$\sum_{k=0}^{N-1} \|a_k\|^2$